

SQL Refinements & Structural Improvements

1. **NULL Handling in Window Functions:** Query 1 (`NTILE() OVER (ORDER BY last_transaction_date ASC)`) places customers with NULL dates (inactive/zero transactions) at the top of the ordering window, distorting Recency scoring. The refined query pushes NULL values to the tail via `NULLS LAST` and filters unlinked records before ranking.
2. **Division by Zero Protection:** Queries 2, 3, 4, 5, and 8 rely on direct division for percentage calculations (`COUNT` or `SUM` denominators). Wrapping all denominators in `NULLIF(..., 0)` prevents division-by-zero crashes on zero-volume periods or filtered subsets.
3. **Data Type Standardization:** In Query 4, `CAST(transaction_datetime AS TIMESTAMP)` within `DATE_TRUNC()` can trigger redundant type casting across entire table scans. The query has been refactored to optimize date grouping.
4. **DRY (Don't Repeat Yourself) Code Efficiency:** Query 7 duplicated a multi-line `CASE` statement across both the `SELECT` and `GROUP BY` clauses. This is refactored using a CTE to project the tier first, eliminating redundant logic evaluations.
5. **Cross-Join Optimization:** Query 6 performed a unfiltered `CROSS JOIN` across full transaction tables before applying standard deviation logic. Standardizing this into a scoped CTE improves execution speed.

Power BI

Section / Page	Metric / Feature	Current Value	Verification Status
Page 1: Overview	Total Transaction Volume	₦560,477,354.85	Verified
Page 1: Overview	Risk Review Trend	19.60 % Avg	Verified
Page 2: Customers	Active Customer Ratio	91.13% (1,367 Users)	Verified
Page 2: Customers	Customer Detail Table	1,500 Records	Verified
Page 3: Transactions	Total Failed Volume	₦ 24,825,751.10	Verified
Page 3: Risk	Channel Failure Rate	5.25%	Verified

Final Validation Evidence Matrix

Finding	Analytical Area	Pre-Validation Premise	Post-Validation Corrective Adjustment
1	Digital Volume	Gross volume drives channel success.	Replaced with Net Settled Volume after finding 8.86% in un-settled failures/reversals.
2	Customer CLV	High frequency = High customer value.	Updated CLV to RFM & Net Margin after identifying high-frequency, low-yield users.
3	Revenue Trajectory	Aggregated quarterly growth is stable.	Added Weekly/Monthly MoM Tracking to catch mid-quarter volume dips (₦2.3M drop in Feb).
4	Risk Distribution	Fraud occurs uniformly across cohorts.	Implemented Cohort-Specific Risk Rules targeting recent digital onboarding pipelines.
5	Channel Efficiency	Speed and low fee = High efficiency.	Adopted Risk-Adjusted Efficiency Scorecard accounting for chargebacks and reversals.